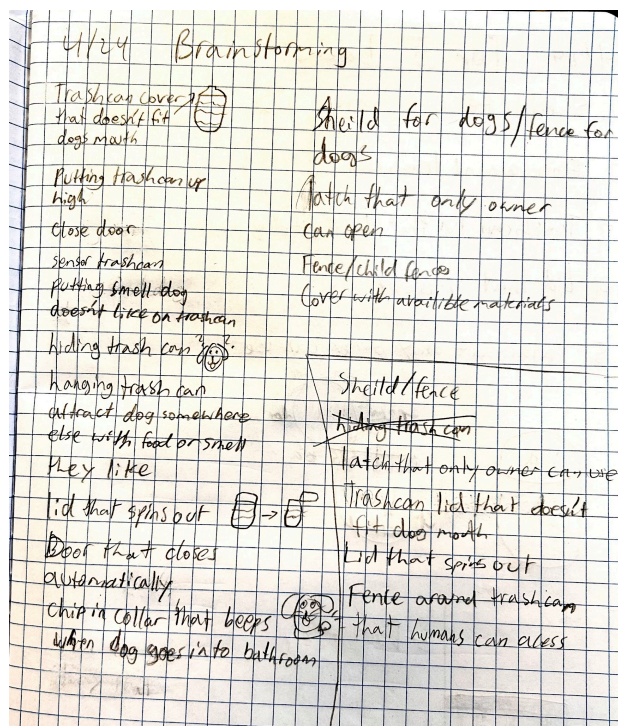


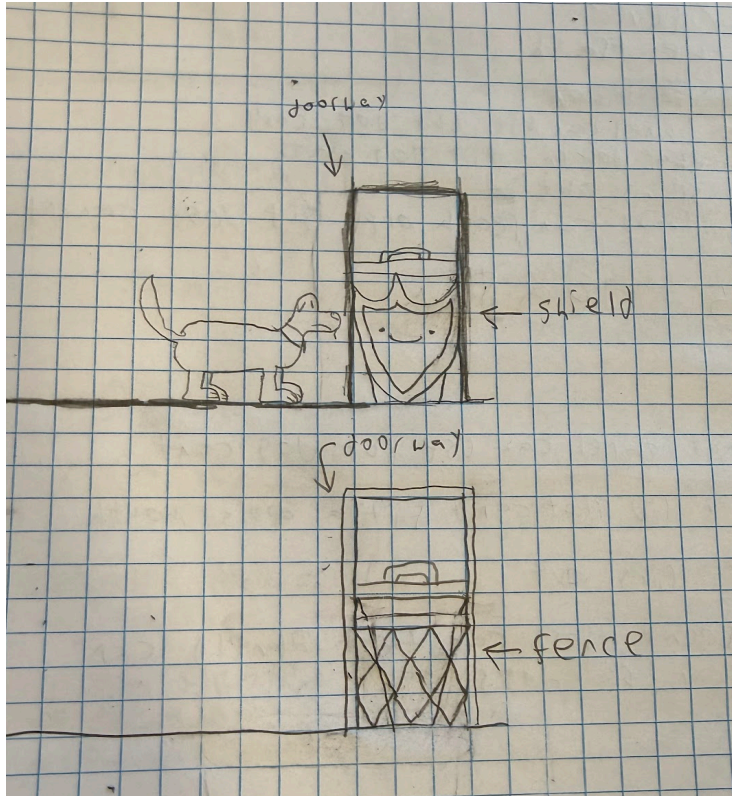
Element D: Design Concept Generation, Analysis, and Selection

Dogs are getting into indoor trash cans, making messes, creating health risks, and causing inconvenience for dog owners and people inside the home. This problem involves dog owners, the dogs themselves, and other people inside the home. This happens inside the house nearly every day. Dogs dig in the trash because they are attracted to the food and smells inside of the trash cans. This is a small but persistent issue that leaves owners irritated.

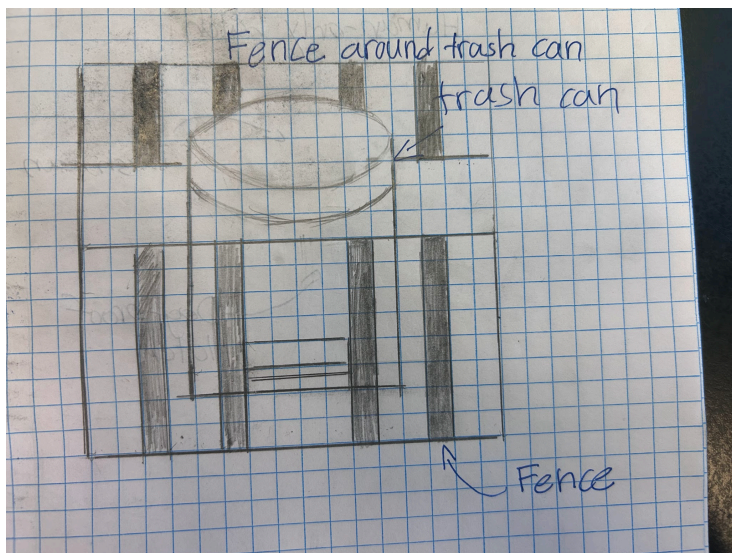
First, we wrote down any ideas that we had in our notebooks for 10 minutes, then we talked with our teammates and wrote down any more ideas. We then chose our five best ideas and put them into a decision matrix to determine the best solution. The stakeholder's current solution was compared to our top five ideas by determining if our top five ideas meet the design requirements better or worse than the stakeholder's current solution.



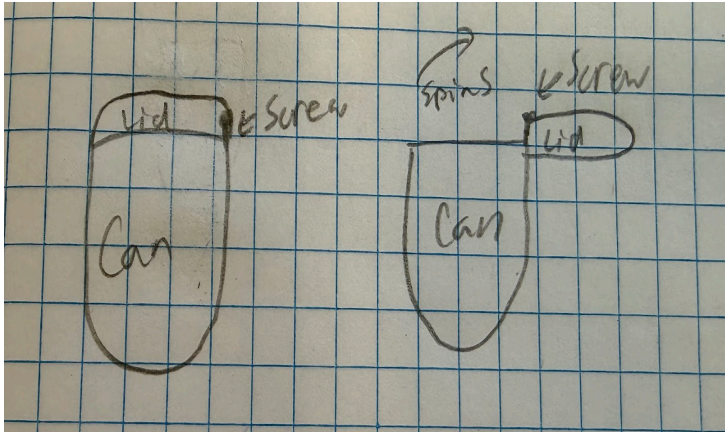
(Brainstorming process)



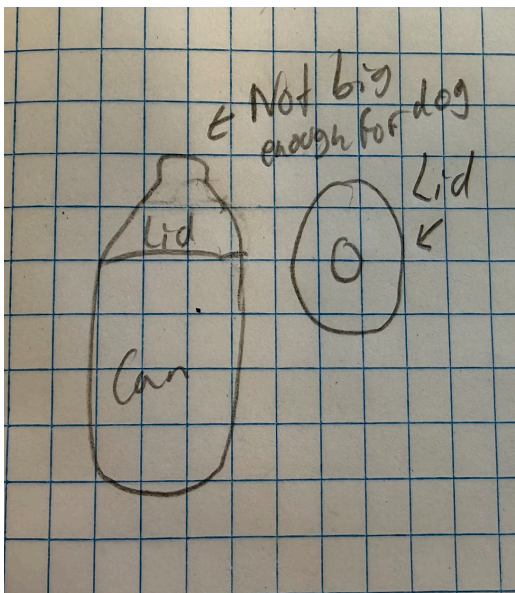
(Idea #1: fence in doorway)



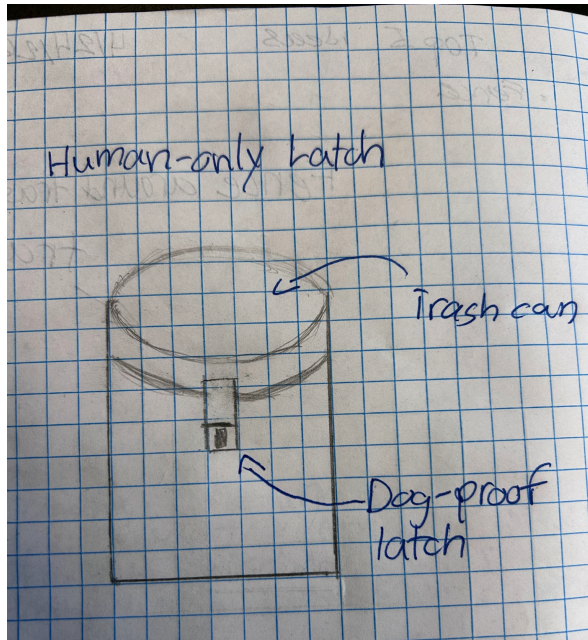
(Idea #2: fence around trashcan)



(Idea #3: lid that spins out)



(Idea #4: lid that does not fit a dog's mouth)



(Idea #5: dog-proof latch)

Our final solution is an elastic cover with a hole in the middle that goes over the trash can. The elastic cover requires few materials to make and is realistic to build. To compare the elastic cover to our other top ideas we used a weighted design matrix. The design matrix helped us determine if our solutions met the design requirements and if they performed them better or worse than the stakeholder's current solution. Safety is weighted double because it is our number one priority to not cause any harm to dogs or owners. Preventing dogs from ripping up the garbage in the trash cans and being universal for all trash cans are also weighted double. Preventing dogs from ripping up the garbage is weighted extra because it is the main problem we are trying to solve and universality is important because we plan to make a one size fits all solution.

	Proposed Solutions					
Design Requirements	Current Solution: Closing Door	Idea #1: Fence in doorway/hallway	Idea #2: Fence Around Trash Can	Idea #3: Lid that Spins Out	Idea #4: Elastic Cover	Idea #5: Dog-proof latch
Does not harm dogs or owners (High) x2	0	+2	+2	0	+2	+2
Prevents dogs from getting to or ripping up trash bags (High) x2	0	0	0	0	0	0
Durable enough to prevent dogs from breaking the device (High)	0	-1	-1	0	0	+1
Universal for all trash cans (High) x2	0	0	0	0	+2	-2
Works for dogs all sizes (High)	0	0	-1	0	-1	+1
Must cost \$20 or less to make (High)	0	0	0	0	0	0
One hand is able to operate trash can (Medium)	0	0	0	0	0	-1
Does not require specialty tools to install (Medium)	0	0	-1	-1	0	-1
Installs in 10 minutes or less (Medium)	0	0	0	0	0	0
Available in multiple colors (Low)	0	-1	-1	+1	+1	+1
Total Score	0	0	-2	2	2	1

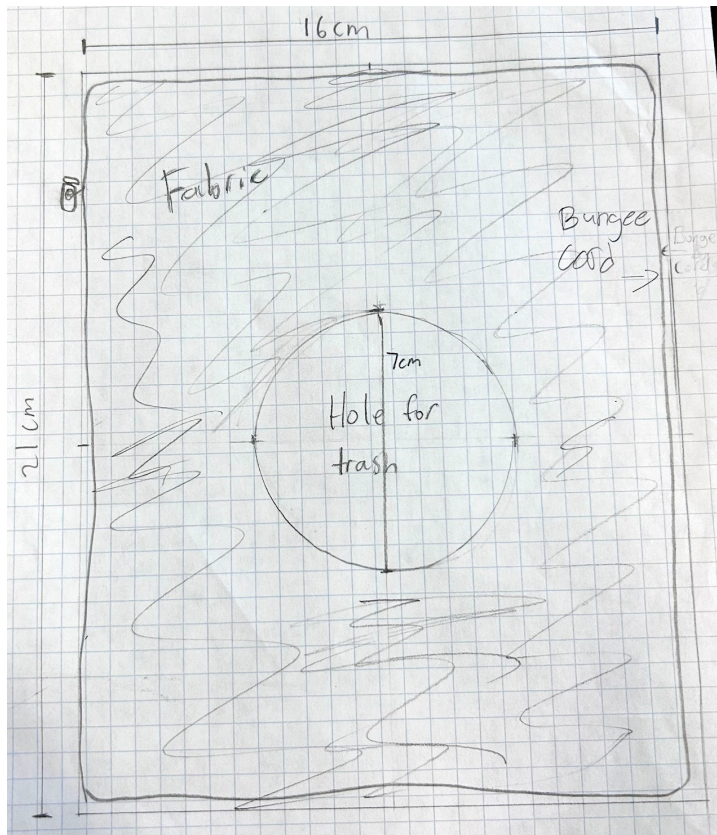
Scoring

0 = same as current solution

+1 = better than current solution

-1 = worse than current solution

Our final solution is a cover with a hole in the middle. The cover will have an elastic band around the outside so that the cover is universal for all trashcans. We chose the elastic cover because it tied for scoring the highest points. We chose the elastic cover because it met the most design requirements and was the most realistic to build.



(Final sketch)

To build our prototype we will cut out a rectangle of outdoor fabric to 45 by 60 centimeters then we cut out a seven centimeter in diameter circle in the middle so that the user will still be able to throw away small pieces of trash like tissue. Once the fabric

has been cut, we will sew a hem that will allow us to thread a bungee cord through the fabric. After the prototype is finished we will run it through a series of tests to make sure if the solution meets all of the design requirements. We were originally going to make our prototype 21 by 16 centimeters, but one of our experts, known as Ms. Z, told us that our design would be too small and would not be able to fit all trash cans, so we decided to change the size of our design to 45 by 60 centimeters based off of a giant bin that was a little bigger than an average bathroom trash can.

The final solution is a cover that fits around a trashcan to cover the opening to prevent dogs from digging in trash cans that do not have lids. The cover will still have a small hole in the middle that can be used to throw away small things while also preventing dogs from digging into the trashcan. The materials and tools that we will be using are outdoor fabric, adjustable bungee cords, thread, tailor's chalk, scissors, and needles.